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Security in Computing and Information Technology
COSC2536 (Semester 2, 2026)
Assignment 2

Assessment Type	To be attempted in a group of 2. **If you want to join a group or change your earlier group, please contact the lecturer by the end of week 9.** Submit online via Canvas→ Assignments→ Assignment 2. Undergraduate [UG] Version
Due Date	Week 11, Sunday 11th October 2026, 11:00 pm *AEST*
Marks	45

1. Overview (you must read this first)

The objective of Assignment 2 is to evaluate your knowledge on the topics covered mainly in Lecture 5 to 10. Topics include Privacy-preserving computations based on RSA, ElGamal, and Paillier Cryptosystems; Digital Signature, Blockchain and Cryptocurrency, Digital Authentication & Steganography. You are required to write code in Python programming language. For calculation and report (Part A & C) you are required to submit a single pdf. Upload one single zipped archive (Python codes and pdf) in Canvas.

The tasks are divided into *three* parts:

Part A: Privacy Preserving Secure Models

Part B: Python Programming

Part C: Analytical

If you find a specification open to interpretation, post a query identifying the specification in the corresponding discussion board for assignment 2. Technological projects/problems in real life do not come with a definitive roadmap, or flowcharts complete with instructions. Often, it is the job of the analyst to clarify the client's requirements. For this assignment and course, the lecturer Kashif Khan is considered the client.

2. Learning Outcomes

This assessment relates to the following learning outcomes of the course:

1. CLO 1: explain the functioning of security services in computing environments and the security issues in networked applications.
2. CLO 2: discuss various types of data integrity and confidentiality mechanisms including public key cryptography.
3. CLO 3: describe basic system security mechanisms and protocols, such as those used in operating systems, file systems and computer networks.
4. CLO 4: analyze the overarching importance of IT security in areas such as networking, databases, operating systems, and web systems.
5. CLO 5: apply privacy principles in basic practical settings in IT environments.
6. CLO 6: analyze and evaluate the security of computing and IT systems on a practical level and privacy related issues in computing.

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3. IMPORTANT INSTRUCTIONS- *please read carefully*

Please follow these instructions carefully:

a. You must write all the code in `Python` programming language-ONLY.

b. You are allowed to use ANY of these Python packages for cryptographic purposes- `cryptography`, `rsa` and `pycryptodome`, `pycryoptodomex`.

c. You must ensure the project is organised inside the `assignment2/` folder. The folder structure should be as follows:

   - Input Files: all files that need to be encrypted should be placed inside the `input/` sub-folder within the main project directory.
   - Encryption Keys: store all the encryption keys inside the `keys/` sub-folder.
   - Output Files: all encrypted and decrypted output files must be saved inside the `output/` sub-folder.

d. You must thoroughly comment all the python files.

e. You must adhere to the naming conventions for the python files, as explained in the individual tasks.

f. You should use Visual Studio Code for programming purposes.

g. You must indent python files professionally. You are advised to download Python programming extension in Visual Studio code that will assist you in writing neat and professional code that follows PEP8 guidelines (https://peps.python.org/pep-0008/).

h. You should not generate and copy code generated verbatim by ChatGPT or any of the AI code generators. IF THE MARKER FEELS that code/content has been ChatGPT or AI generated, you will be asked to attend a face-to-face demo and asked to explain your code/content. Post demo, such cases may be referred for academic integrity hearing. The hearing details remain on the permanent student record. Please be careful.

i. Hand-written answers are not allowed and will not be assessed. Compose your answers using any word processing software (e.g., MS Word).

j. You are required to show all of the steps, intermediate results and screenshots for each question with calculation or where necessary. No supporting explanation, calculations and screenshots will lead to NO MARKS.

k. You may use online websites for mathematical calculations.

Please go to next page for the tasks →

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4. Questions/Tasks

Download the zipped file based on your operating system, it contains the files needed for the following tasks.

Part A: Privacy Preserving Secure Models (Marks: 5)

Privacy Preserving Online Voting System

Recently, several controversies have been observed in voting around the world. Even electronic voting can be manipulated[1]. In an electronic voting system, the voting authority cannot be trusted completely as it can be biased. Using a privacy-preserving online voting system removes controversy in the voting system. In this privacy-preserving online voting system, voters encrypt their votes in the voting booth before sending them to the voting authority. A voting server records each encrypted vote and determines the voting result on behalf of the voting booth as the voting booth does not have enough computation power. The encrypted result is sent to the voting authority, which determines the winner based on encrypted votes. The homomorphic property of the Paillier Cryptosystem allows the addition of encrypted numbers. Therefore, the Paillier Cryptosystem can be used in developing a privacy-preserving online voting application.

In this task, you need to design a privacy preserving online voting system as shown in **Figure-1** using the **Paillier cryptosystem.** Suppose that a group of students want to elect their club president.

Your designed privacy-preserving voting application must fulfil the following requirements:

- Votes must be **encrypted** from **Voting Booth** using **Paillier Cryptosystem** before sending them to the Voting Server.
- A vote is an integer number which should be equivalent to a **12-bit binary string**.
- The number of candidates should be three. For example, **ALICE, BOB, and EVE**.
- There are **10 voters** who will take part in the voting**.**
- Assume that **4 voters** will vote for **ALICE**, **3 voters** will vote for **BOB,** and **3 voters** will vote for **EVE**. After counting the votes, the **Voting Authority (VA)** should find **4 votes for Alice, 3 votes for BOB,** and **3 votes for EVE**.
- The **Voting Authority** chooses ***p=107***, ***q=61*** and select ***g=7019***.
- The private numbers chosen by **10 voters** and their votes are as follows:

Table-1: Votes of 10 voters

Voter No.	Voter's Private Number, *r*	Vote for	Voting message, *m*
1	*r1=71*	ALICE	000100000000 = 256
2	*r2=72*	ALICE	000100000000 = 256
3	*r3=73*	ALICE	000100000000 = 256
4	*r4=74*	ALICE	000100000000 = 256
5	*r5=75*	BOB	000000010000 = 16
6	*r6=76*	BOB	000000010000 = 16
7	*r7=77*	BOB	000000010000 = 16
8	*r8=78*	EVE	000000000001 = 1
9	*r9=79*	EVE	000000000001 = 1
10	*r10=80*	EVE	000000000001 = 1

- The **Voting Authority** sets up required **public and private keys** and makes the public-keys to all **voting booth** before the vote starts.

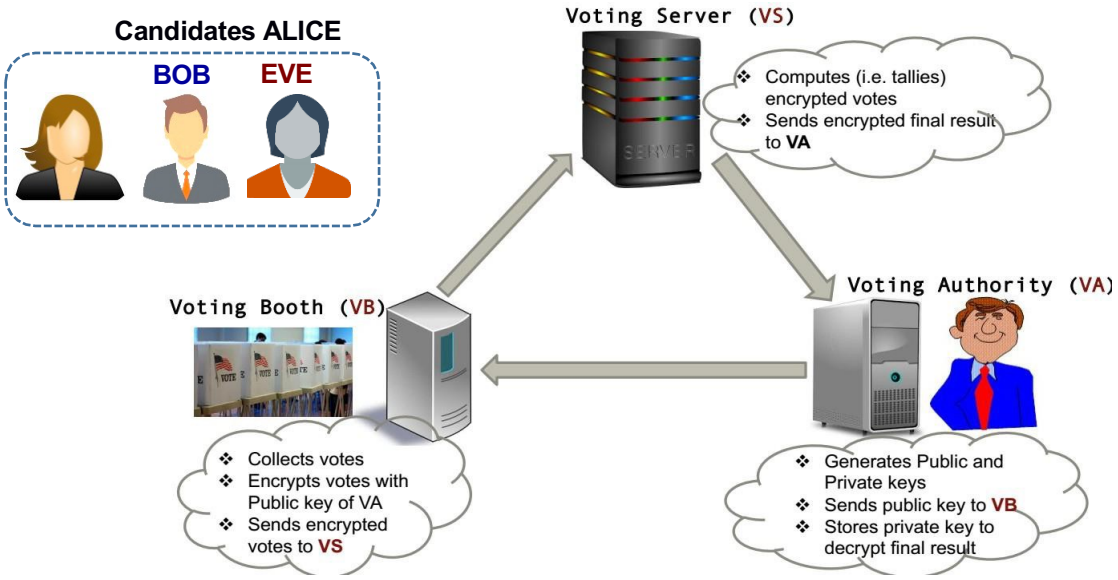


Figure-1: Privacy-preserving voting system

Q1 (5 marks) Using the provided values of ***p, q, g,*** and **Voter's Private Numbers** (values of r1, r2, …, r12 as shown in **Table-1**), show detail step-by-step computations of the **key generation**, **encryption**, **homomorphic computations**, **and decryption** processes for votes of **10 voters.**

[1.https://www.dw.com/en/democracy-in-danger-elections-are-easy-to-manipulate/a-45858161]

Part B: Python Programming (Marks: 5+5+10)

Q.2 (5 marks) Write a program in python (task2-rsa-manual.py) to manually implement to encrypt and decrypt a string message (your student number without the s). You are **NOT** allowed to use any python cryptographic libraries.

In other words, the task involves generating large prime numbers, computing public and private keys, and using those keys to encrypt and decrypt messages. All mathematical calculations must be done by Python **without** the use of cryptographic libraries. **All the file paths must use BASE variable to make the code work on all operating systems**.

Q 3 (5 marks) Write a program in python (task3-hybrid.py) that uses hybrid encryption to secure the exchange of the file (task2.txt) with your friend. The program will use hybrid technique- symmetric **(AES)** and asymmetric encryption (**RSA)** to make the communication more **secure**.
Requirements: The program must display all the keys, encrypted and decrypted outputs to the user. The RSA keys and decrypted file must be stored **in separate files. Textbook RSA is not allowed.**

All the file paths must use BASE variable to make the code work on all operating systems.

Any suspicious looking code will flag the marker, and we will invite you for a face-to-face demo.

Q4. (5 Marks) Encrypted messages can indeed be hidden inside an image using a technique called **steganography**. Steganography allows cybersecurity professionals to conceal information within other, seemingly innocuous data, such as images, audio files, or even text. When applied to images, this process involves embedding encrypted or hidden data into the image in such a way that the image itself appears unchanged to the human eye.

Write a program in python (task4-hide-and-seek.py) that encrypts a message using **AES** and embeds the message in a jpeg file using LBS or metadata steganography techniques. The program is then able to decrypt and reveal this secret message as well.

All the file paths must use BASE variable to make the code work on all operating systems.

Q 5 **Video Presentation [5 Marks]**

Prepare a video presentation explaining and demonstrating your solution.
The presentation should include but not limited to:

- the purpose of the program;
- how AES encryption is used;
- how the encrypted message is hidden inside the image;
- how the hidden message is extracted and decrypted;
- demonstration of the program using your own test data; and
- the main strengths and limitations of your approach.

The presentation should demonstrate your understanding of the implementation rather than simply reading the code or report.

Part C: Analytical [Marks: 10+10]

Q6. Report Writing on Secure or Privacy-Preserving Systems (Marks: 10)

PART A) **[5 Marks]**

This question requires you to investigate a legacy system of your own choice and architect a proposal for its modernization using blockchain technology. The goal is to understand the shortcomings and how decentralized principles can create more secure, transparent, and efficient solutions.

The report must contain the followings (but not limited to):

i. Based on the topic you have chosen, give overview of the existing systems with one or more scenarios to illustrate the requirement of the privacy preserving techniques or blockchain (i.e., **significance**). Use necessary diagrams.
ii. Discuss some of the existing systems that have already been introduced to address the requirements (i.e., **background**).
iii. Describe what limitations will be addressed in your proposed system (i.e., **aims**) and what new features to be included in your proposed system (i.e., **objectives**).
iv. Describe your proposed systems in detail (i.e., **system overview**) with necessary diagrams
v. Discuss how your proposed system will benefit Australia and other countries technologically, economically, and socially (i.e., **benefits**).
vi. Discuss the strength and limitations of your proposed system compared to existing systems. (i.e., advantages and limitations)
vii. Providing a concluding statement on your proposed systems (i.e., **conclusion**)

The report should be well structured, having a proper significance of the system you have chosen

based on the limitations of existing systems, aims and objectives of your system, detailed discussion of your selected systems, and concluding remarks. You must provide necessary diagrams based on your own thoughts as well as collected from different sources. **Important:** If you select a system that has already been converted to blockchain, your report will receive **no marks**.

Articles can be searched in:

- Google,
- Google Scholar (https://scholar.google.com/),
- IEEEXplore (https://ieeexplore.ieee.org/Xplore/home.jsp), etc.

PART B) **Video Presentation [5 Marks]**

Prepare a video presentation explaining your proposed blockchain-based system. The presentation should include, **but is not limited to**, the problem with the existing legacy system, the proposed architecture, how blockchain is used, and the main benefits and limitations of your solution. The presentation should focus on **your proposed system and design decisions**, rather than simply reading the written report.

Q 7 (Marks: 10)

Scenario:

Alice purchased an inexpensive software package online to establish secure communication with **Bob** over a public network using public-key cryptography. After some time, Alice was informed that an individual named **Trudy** had been able to compromise the supposedly secure communication between Alice and Bob.

You have been hired as a security analyst to investigate the cause of the security breach, test the supplied software, and determine how the weakness can be addressed.

Attached Files:

a) Cipher.exe

This executable generates a public key and ciphertext from information entered by the user.

- The user must enter student id (numbers only), first name and a last name using English alphabetic characters.
- The user must enter two prime numbers. Each prime number must be greater than 1,000,000.
- Individual: the first prime number must be the immediate prime less than your student ID, and the second prime number must be the immediate prime greater than your student ID.
- Group: the first prime number must be the immediate prime greater than Student 1's ID, and the second prime number must be the immediate prime greater than Student 2's ID.
- The user must then enter a plaintext message containing a minimum of 3 English alphabetic characters. Special characters, spaces, and numbers are not allowed.
- Before terminating, it will display a small number of general statements or observations. These statements may or may not be relevant to the security

weakness. Treat them only as possible clues and use evidence from your own testing to determine whether they are significant.

- The executable generates two files:
  - key.txt – containing the public key.
  - cipher.txt – containing the ciphertext and initials of name.

b) Hack.exe

- This executable is designed to attempt to break the encrypted text generated by Cipher.exe.
- Copy key.txt and cipher.txt into the same folder as Hack.exe before running it.

Instructions:

1. Run Cipher.exe and provide the required first name, last name, prime numbers, and plaintext.
2. Observe the program output and retain key.txt and cipher.txt.
3. Copy key.txt and cipher.txt into the folder containing Hack.exe.
4. Run Hack.exe and record whether, and under what conditions, it can recover the plaintext.
5. Repeat tests where necessary. Base your conclusions on reproducible evidence rather than assuming that every message displayed by Cipher.exe is relevant.

Tasks:

Q7.1 [3 Marks]:

Using your analytical skills, systematically test Cipher.exe and Hack.exe and determine what is happening and why the secure communication can be compromised. Support your explanation with evidence from your experiment, including relevant inputs, generated outputs, and observations. Consider all observable program behaviour, including the name inputs and any statements displayed at the end. These statements may contain useful clues, but you must use evidence to determine whether they are relevant

Q7.2 [3 Marks]:

Repeat the investigation using larger prime numbers selected from a concatenated student ID, as follows:

- Group: concatenate Student 1's ID with Student 2's ID. For example, if the IDs are 1234567 and 1122334, the concatenated ID is 12345671122334. Select the required prime numbers according to the Cipher.exe rules for the concatenated value.
- [**IF YOUR INDIVIDUAL SUBMISSION IS APPROVED**]
  Individual: concatenate your student ID with itself. For example, if your student ID is 1234567, the concatenated ID is 12345671234567. Select the immediate prime below and the immediate prime above the concatenated ID in the same manner required by Cipher.exe.

Investigate whether changing the size of the prime numbers changes the behaviour or security of the system. In particular, analyse the relationship, if any, between the selected prime numbers, the plaintext, the resulting ciphertext, and Hack.exe's ability to recover the message. Submit a maximum one-page report

supported by evidence.

Q7.3 [4 Marks]:

Using your programming skills, write a Python program that generates ciphertext which Hack.exe cannot successfully break under your test conditions.

- Your program must accept the same core user inputs as the revised Cipher.exe task: first name, last name, two valid prime numbers, and a plaintext message.
- Your program must generate files named key.txt and cipher.txt so that they can be tested directly with Hack.exe.
- Do not simply hard-code ciphertext for one test case. Your solution should implement a defensible cryptographic approach and work for any valid inputs.
- After creating your Python program, place the generated key.txt and cipher.txt in the same folder as Hack.exe and run Hack.exe to test your solution.
- Briefly explain why your revised approach is expected to resist the weakness identified in Q7.1 and support your claim with test evidence.
- **All the file paths must use BASE variable to make the code work on all operating systems**.

Important Notes:

- If you suspect that Cipher.exe is using key logging or another local mechanism, copy key.txt and cipher.txt to a different computer or folder together with Hack.exe and repeat the analysis.
- Use the operating-system-specific Assignment 2 package supplied through Canvas.
- This task requires initiative, self-research, testing, and application of security concepts. Your report must explain your own observations and conclusions, and you should be prepared to demonstrate and explain your code and analysis.

Video Presentation Requirements

- At the beginning of the video, each student must show their **RMIT Student ID card** on camera and clearly state their **full name and student number**.
- The video must capture both the **student's face and computer screen** simultaneously.
- Students must explain the work in their **own words**, including the reasoning behind the methods, code, tools, and design decisions used. Simply showing code, commands, or outputs without explanation will not receive full marks.
- For group submissions, **each group member must appear in the video and make a meaningful contribution**.
- Upload the video to **Microsoft SharePoint or YouTube** and include a **streamable link** in the assignment report. The link must remain accessible until final grades are released. Zero marks if link is not working (available) at any point in time.
- The presentation should be a **live walkthrough/demonstration** of the submitted work rather than a slide-only presentation.

5. Submission and extension

Zip the whole project. **MAKE SURE THAT** zipped archive file is not corrupted and that it can be unzipped- after you have checked the zipped archive, submit 1 zipped archive via Canvas assignment 2 page. Name the document as

- **sNumber1-sNumber2-assignment2.zip**
- **If individual submission APPROVED**, **sNumber-assignment2.zip**

Don't forget: Each member must individually sign and submit the AI Declaration Form. A late penalty will apply if the form is submitted after the deadline.

"Assignments received late and without prior extension approval or special consideration will be penalised by a deduction of 10% of the total score possible per calendar day late for that assessment."

All extension-related requests must be sent 24 hours prior to the deadline to the lecturer (**kashifhesham.khan@rmit.edu.au**).

Assessment declaration: when you submit work electronically, you agree to the assessment declaration:

https://www.rmit.edu.au/students/student-essentials/assessment-and-results/how-to-submit-your-assessments

6.Academic integrity and plagiarism

Academic integrity is about the honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge and ideas. You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e. directly copied), summarised, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods,
- Provide a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites.

If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct. Plagiarism covers a variety of inappropriate behaviours, including:

- Contract cheating- paying someone to do your work
- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students
- Posting assignment tasks on technical forums (*reddit, stack exchange, etc.*) and asking for solution(s)

For further information on our policies and procedures, please refer to:

https://www.rmit.edu.au/students/student-essentials/assessment-and-results/academic-integrity

7. Use of A1 Tools

In this assessment, you may use AI **(RMIT VAL)** only for surface-level support. Permitted uses include spelling, grammar, punctuation, sentence clarity, formatting, transcription, or approved accessibility support.
You must not use AI to generate ideas, structure, analysis, arguments, code, explanations, design decisions, reflections, or other substantive content.

You must submit the **Condition 2 Limited Surface AI Declaration** with your assessment, **even if you did not use AI. If you did not use AI, you only need to complete the section confirming that no AI was used.**

If you used AI for permitted surface-level support, your declaration must identify the tool used, explain exactly what it was used for, confirm what the AI did not do, and include at least one annotated before-and-after sample.

Your assessment submission is not complete unless it includes both the assessment work and the completed Condition 2 Limited Surface AI Declaration.

If you submit the assessment work without the required declaration, the whole assessment will be treated as incomplete. The submission date for the assessment will be the date on which both the assessment work and the completed declaration are submitted.

Late penalties will apply from the original due date, unless you have an approved extension or special consideration.

Assignments submitted more than 10 calendar days after the due date will not be accepted and will receive a mark of zero. This includes submissions where the assessment work was uploaded earlier but the required declaration was not submitted within 10 calendar days after the due date.

Your declaration must be completed carefully, accurately and in full. It is part of the evidence used to confirm that your submission meets the AI condition for this assessment. If your declaration is incomplete, unclear, inconsistent with the submitted work, or raises concerns about the nature or extent of AI use, you may be asked to attend a follow-up conversation or oral authentication session. Where there is evidence that AI has been used beyond the permitted scope, or that AI use has not been declared accurately, the matter may be referred for academic integrity review.

8. Marking Guidelines

The marks allocated have been added to each of the tasks. **Please read rubrics for detail**